

2.2 Evaluating Determinant by Row Reduction

Property 1:

$$\begin{vmatrix} ka_1 & ka_2 & ka_3 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k \begin{vmatrix} a_1 & a_2 & a_3 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\det(B) = k \det(A)$$

In matrix B the first of A was multiplied by k.

Property 2

$$\begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\det(B) = -\det(A)$$

In matrix B first & second rows of A were interchanged.

Property 3

$$\begin{vmatrix} a_{11} + ka_{21} & a_{12} + ka_{22} & a_{13} + ka_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\det(A) = \det(B)$$

In matrix B a multiple of second row of A was added to first row.

Property 4

If

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

$$= 3 \begin{vmatrix} 1 & 2 & 1 \\ 4 & 5 & 2 \\ 7 & 8 & 3 \end{vmatrix}$$

Property: 5

If A is square matrix with two proportional row or two proportional columns, then $\det(A) = 0$.

Example: 1 (Determinants of Elementary Matrices)

$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$	$= 3$	$\begin{vmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}$	$= -1$
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Second row of I_4 is multiplied by 3.

First and last rows of I_4 were interchanged.

$$\begin{vmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

7 times the last of I_4 was added to first row.

$$\det(A) = (1)(7)(3)(-26)$$

$$\det(A) = -546 \text{ Ans}$$

Example: 5 (Row operations & Cofactor Expansion)

$$A = \begin{bmatrix} 3 & 5 & -2 & 6 \\ 1 & 2 & -1 & 1 \\ 2 & 4 & 1 & 5 \\ 3 & 7 & 5 & 3 \end{bmatrix}$$

$$\det(A) = \begin{bmatrix} 3 & 5 & -2 & 6 \\ 1 & 2 & -1 & 1 \\ 2 & 4 & 1 & 5 \\ 3 & 7 & 5 & 3 \end{bmatrix}$$

$$\det(A) = \begin{bmatrix} 0 & -1 & 1 & 13 \\ 1 & 2 & -1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 1 & 8 & 0 \end{bmatrix} \begin{array}{l} R_1 - 3R_2 \\ R_3 - 2R_2 \\ R_4 - 3R_2 \end{array}$$

Expand along 1st Col:

$$\det(A) = 0 + (-1)^{2+1} \begin{vmatrix} -1 & 1 & 3 \\ 0 & 3 & 3 \\ 1 & 8 & 0 \end{vmatrix} + 0 + 0$$

$$\det(A) = - \begin{vmatrix} -1 & 1 & 3 \\ 0 & 3 & 3 \\ 1 & 8 & 0 \end{vmatrix}$$

Expand along 2nd row

$$\det(A) = - \left[0 + (-1)^{2+1} (3) \begin{vmatrix} -1 & 3 \\ 1 & 0 \end{vmatrix} + (-1)^{2+3} (3) \begin{vmatrix} -1 & 1 \\ 1 & 8 \end{vmatrix} \right]$$

$$\det(A) = - \left[0 + 3(0-3) - 3(-8-1) \right]$$

$$\det(A) = - (-9 + 27) = -18$$

$$\det(A) = -18 \text{ Ans}$$

Exercise : 2.2

Question : 1-4

Verify that $\det(A) = \det(A^T)$

Use given matrix and verify it
it's easy.

Question : 5-8

Find determinant of elementary matrix
by inspection.

5-

$$E_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Third row of T_1 was multiplied by -5

→ If row is multiplied by k the det. is k

→ So $\det(E) = -5$

6-

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 5R_1$$

→ Adding a multiple of one row to another does't change deter.

→ So $\det(E) = 1$

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$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} \rightarrow R_2 \leftrightarrow R_3 \\ \rightarrow \text{Swapping two rows} \\ \text{multiplies det by } -1 \\ \rightarrow \text{So } \det(E) = -1 \end{array}$$

8-

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

R_2 of I_4 was multiplied by $-\frac{1}{3}$

So $\det(E) = -\frac{1}{3}$

Question: 9-14

Evaluate determinant of given matrix by first reducing matrix to row echelon form and then using some combination of row operation and cofactor expansion.

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$$A = \begin{bmatrix} 3 & -6 & 9 \\ -2 & 7 & -2 \\ 0 & 1 & 5 \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} 3 & -6 & 9 \\ -2 & 7 & -2 \\ 0 & 1 & 5 \end{vmatrix}$$

$$\det(A) = 3 \begin{vmatrix} 1 & -2 & 3 \\ -2 & 7 & -2 \\ 0 & 1 & 5 \end{vmatrix} \rightarrow \text{common } 3$$

$$\det(A) = -3 \quad \begin{vmatrix} 1 & -2 & 3 \\ 0 & 3 & 4 \\ 0 & 1 & 5 \end{vmatrix} \quad R_2 + 1R_1$$

$$\det(A) = -3 \quad \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 3 & 4 \end{vmatrix} \quad R_2 \leftrightarrow R_3$$

$$\det(A) = -3 \quad \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & -11 \end{vmatrix} \quad R_3 - 3R_2$$

Expand along C_1

$$\det(A) = -3 \left[(-1)^{1+1} (1) \begin{vmatrix} 1 & 5 \\ 0 & -11 \end{vmatrix} + 0 + 0 \right]$$

$$\det(A) = -3 \left[1(-11 - 0) \right]$$

$$\det(A) = 33 \text{ Ans}$$

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$$\begin{vmatrix} 3 & 6 & -9 \\ 0 & 0 & -2 \\ -2 & 1 & 5 \end{vmatrix}$$

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$$\det(A) = \begin{vmatrix} 3 & 6 & -9 \\ 0 & 0 & -2 \\ -2 & 1 & 5 \end{vmatrix}$$

$$\det(A) = 3 \begin{vmatrix} 1 & 2 & -3 \\ 0 & 0 & -2 \\ -2 & 1 & 5 \end{vmatrix}$$

$$\det(A) = 3 \begin{vmatrix} 1 & 2 & -3 \\ 0 & 0 & -2 \\ 0 & 5 & -1 \end{vmatrix} \quad R_3 + 2R_1$$

$$\det(A) = -3 \begin{vmatrix} 1 & 2 & -3 \\ 0 & 5 & -1 \\ 0 & 0 & -2 \end{vmatrix} \quad R_2 \leftrightarrow R_3$$

$$\det(A) = -3(5) \begin{vmatrix} 1 & 2 & -3 \\ 0 & 1 & -\frac{1}{5} \\ 0 & 0 & -2 \end{vmatrix}$$

$$\det(A) = -3(5)(-2) \begin{vmatrix} 1 & 2 & -3 \\ 0 & 1 & -\frac{1}{5} \\ 0 & 0 & 1 \end{vmatrix}$$

Expand along 3rd row

$$\det(A) = 30 \left[0 + 0 + (-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \right]$$

$$\det(A) = 30 \text{ Ans}$$

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2	1	3	1
1	0	1	1
0	2	1	0
0	1	2	3

$R_1 \leftrightarrow R_2$
↑

$$\det(A) = \begin{vmatrix} 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix} \rightarrow \det(A) = - \begin{vmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 3 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix}$$

$$\det(A) = - \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix} \quad R_2 - 2R_1$$

$$\det(A) = - \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 1 & 4 \end{vmatrix} \quad \begin{array}{l} R_3 - 2R_2 \\ R_4 - R_2 \end{array}$$

$$\det(A) = + \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -1 & 2 \end{vmatrix} \quad R_3 \leftrightarrow R_4$$

$$\det(A) = \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 6 \end{vmatrix} \quad R_4 + R_3$$

$$\det(A) = 6 \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Expand along 4th Row

$$\det(A) = 6 \left[0 + 0 + 0 + (-1)^{4+4} \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} \right]$$

$$\det(A) = 6 \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

Again expand along 3rd Row

$$\det(A) = 6 \left[0 + 0 + (-1)^{3+3} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \right]$$

$$\det(A) = 6 \text{ Ans}$$

$$\det(A) = -17 \text{ Ans}$$

13-

1	3	1	5	3
-2	-7	0	-4	2
0	0	1	0	1
0	0	2	1	1
0	0	0	1	1

$$\det(A) =$$

1	3	1	5	3
-2	-7	0	-4	2
0	0	1	0	1
0	0	2	1	1
0	0	0	1	1

$$\det(A) =$$

1	3	1	5	3
0	-1	2	6	8
0	0	1	0	1
0	0	2	1	1
0	0	0	1	1

$$R_2 + 2R_1$$

$$\det(A) =$$

$$-1$$

1	3	1	5	3
0	1	-2	-6	-8
0	0	1	0	1
0	0	2	1	1
0	0	0	1	1

$$-R_2$$

$\det(A) = -1$	1	3	1	5	3	
	0	1	-2	-6	-8	$R_1 - 2R_3$
	0	0	1	0	1	
	0	0	0	1	-1	
	0	0	0	1	1	

$\det(A) = -1$	1	3	1	5	3	
	0	1	-2	-6	-8	$R_5 - R_4$
	0	0	1	0	1	
	0	0	0	1	-1	
	0	0	0	0	2	

$\det(A) = -2$	1	3	1	5	3	
	0	1	-2	-6	-8	
	0	0	1	0	1	
	0	0	0	1	-1	
	0	0	0	0	1	

Expand along 5th Row.

$\det(A) = -2$										
	0	+	0	+	0	+	0	+	⁵⁺⁵ $(-1)(1)$	$\begin{vmatrix} 1 & 3 & 1 & 5 \\ 0 & 1 & -2 & -6 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$

$$\det(A) = -2 \quad \begin{vmatrix} 1 & 3 & 1 & 5 \\ 0 & 1 & -2 & -6 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

again Expand along 4th Row.

$$\det(A) = -2 \quad \begin{vmatrix} & & & 4+4 & \begin{vmatrix} 1 & 3 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{vmatrix} \\ 0 & + & 0 & + & 0 & + & (-1)(1) \end{vmatrix}$$

$$\det(A) = -2 \quad \begin{vmatrix} 1 & 3 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{vmatrix}$$

$\det(A) = -2$ Ans

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$$\begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{vmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -3 \\ 2 & 8 & 6 & 1 \end{vmatrix}$$

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$$\det(A) = \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 12 & 0 & -1 \end{vmatrix} \begin{array}{l} R_2 - 5R_1 \\ R_3 + R_1 \\ R_4 - 2R_1 \end{array}$$

$$\det(A) = \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 108 & 23 \end{vmatrix} R_4 - 12R_2$$

$$\det(A) = -3 \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 108 & 23 \end{vmatrix}$$

$$\det(A) = -3 \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & -13 \end{vmatrix} R_4 - 108R_3$$

$$\det(A) = 3(-13) \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Expand along 4th Row

$$\det(A) = 39 \left[0 + 0 + 0 + (-1)(1) \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & -9 \\ 0 & 0 & 1 \end{vmatrix} \right]$$

$$\det(A) = 39 \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & -9 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\det(A) = 39 \left[1 \cdot 1 \cdot 1 + 0 + 0 + (-1) \cdot 1 \cdot 1 \right] \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix}$$

$$\det(A) = 39 \text{ Ans}$$

Question : 15-22

Evaluate Determinant give that

a	b	c	
d	e	f	= -6
g	h	i	

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d	e	f	
g	h	i	= ?
a	b	c	

Given:

a	b	c	
d	e	f	= -6
g	h	i	

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$$\begin{vmatrix} d & e & f \\ a & b & c \\ g & h & i \end{vmatrix} = 6 \text{ Swapped}$$

$$\begin{vmatrix} d & e & f \\ g & h & i \\ a & b & c \end{vmatrix} = -6 \text{ Ans swapped}$$

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$$\begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix}$$

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$$

$$\begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_3} = 6 \text{ Ans}$$

17-

$$\begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix}$$

$$\begin{array}{ccc|c} a & b & c & \\ d & e & f & -6 \\ g & h & i & \end{array}$$

$$3R_1 \quad \begin{array}{ccc|c} 3a & 3b & 3c & \\ d & e & f & -6(3) \\ g & h & i & \end{array}$$

$$4R_3 \quad \begin{array}{ccc|c} 3a & 3b & 3c & -18(4) \\ d & e & f & \\ 4g & 4h & 4i & \end{array}$$

$$-R_2 \quad \begin{array}{ccc|c} 3a & 3b & 3c & \\ -d & -e & -f & = 72 \text{ Ans} \\ 4g & 4h & 4i & \end{array}$$

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$$\begin{array}{ccc|c} a+d & b+e & c+f & \\ & -e & -h & \\ g & h & i & \end{array}$$

$$\begin{array}{ccc|c} a & b & c & \\ d & e & f & -6 \\ g & h & i & \end{array}$$

$$R_1 + R_2 \quad \begin{array}{ccc|c} a+d & b+e & c+f & \\ d & e & f & -6 \\ g & h & i & \end{array}$$

$$-R_2 \left| \begin{array}{ccc|c} a+d & b+e & c+f & \\ -d & -e & -f & \\ g & h & i & \end{array} \right| = +b \text{ Ans}$$

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$$\left| \begin{array}{ccc|c} a+g & b+h & c+i & \\ d & e & f & \\ g & h & i & \end{array} \right|$$

$$\left| \begin{array}{ccc|c} a & b & c & \\ d & e & f & \\ g & h & i & \end{array} \right| = -b$$

$$R_1 + R_3 \left| \begin{array}{ccc|c} a+g & b+h & c+i & \\ d & e & f & \\ g & h & i & \end{array} \right| = -b \text{ Ans}$$

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$$\left| \begin{array}{ccc|c} a & b & c & \\ 2d & 2e & 2f & \\ g+3a & h+3b & i+3c & \end{array} \right|$$

$$\left| \begin{array}{ccc|c} a & b & c & \\ d & e & f & \\ g & h & i & \end{array} \right| = -b$$

$$2R_2 \quad \begin{array}{ccc|c} a & b & c & \\ \hline 2d & 2e & 2f & = (-6)(2) \\ g & h & i & \end{array}$$

$$R_3 + 3R_1 \quad \begin{array}{ccc|c} a & b & c & \\ \hline 2d & 2e & 2f & = -12 \text{ Ans} \\ g+3a & h+3b & i+3c & \end{array}$$

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$$\begin{array}{ccc|c} -3a & -3b & -3c & \\ d & e & f & \\ g-4d & h-4e & i-4f & \end{array}$$

$$\begin{array}{ccc|c} a & b & c & \\ d & e & f & = -6 \\ g & h & i & \end{array}$$

$$-3R_1 \quad \begin{array}{ccc|c} -3a & -3b & -3c & = (-6)(-3) \\ d & e & f & \\ g & h & i & \end{array}$$

$$R_3 - 4R_2 \quad \begin{array}{ccc|c} -3a & -3b & -3c & \\ d & e & f & = +18 \text{ Ans} \\ g-4d & h-4e & i-4f & \end{array}$$

22-

a	b	c
d	e	f
2a	2b	2c

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$$

a	b	c
d	e	f
2a	2b	2c

As R_1 & R_3 are proportional so determinant will be 0.

Question: 23

Use Row Reduction to show

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$$

Taking Transpose

$$= \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix} \begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \end{array}$$

$$= \begin{vmatrix} 1 & a & a^2 \\ 0 & (b-a) & (b-a)(b+a) \\ 0 & (c-a) & (c-a)(c+a) \end{vmatrix}$$

$$= (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 1 & c+a \end{vmatrix}$$

$$= (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+a \\ 0 & 0 & (c+a) - (b+a) \end{vmatrix} \begin{array}{l} R_3 - R_2 \end{array}$$

Triangular matrix:

$$= (b-a)(c-a)(c-b)$$

so verified

Question: 23

verify formula in part (a) and (b) and make conjecture about a general result of which these are special cases.

a-

$$\det \begin{vmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = -a_{13}a_{22}a_{31}$$

Expand along 1st row.

$$= 0 + 0 + (-1)^{1+3} a_{13} \begin{vmatrix} 0 & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$= a_{13} (0 - a_{31} a_{22})$$

$$= -a_{13} a_{22} a_{31}$$

verified

b-

$$\det \begin{vmatrix} 0 & 0 & 0 & a_{14} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = a_{14} a_{23} a_{32} a_{41}$$

Expand along 1st row.

$$= 0 + 0 + 0 + (-1)^{1+4} a_{14} \begin{vmatrix} 0 & 0 & a_{23} \\ 0 & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{vmatrix}$$

$$= -a_{14} \begin{vmatrix} 0 & 0 & a_{23} \\ 0 & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{vmatrix}$$

$$= -a_{14} \left[0 + 0 + (-1)^{1+3} a_{23} \begin{vmatrix} 0 & a_{32} \\ a_{41} & a_{42} \end{vmatrix} \right]$$

$$= -a_{14}a_{23} (0 - a_{41}a_{32})$$

$$= a_{14}a_{23}a_{32}a_{41}$$

verified

Q: What does special case means?

It means these two examples are small example of bigger general rule.

General result (conjecture)

For this type of matrix, det. equals

- Product of anti-diagonal entries
- With sign depending on matrix size

$$\det(A) = (-1)^{\frac{n(n-1)}{2}} \left(\text{Product of anti diagonal entries} \right)$$

n = matrix size

Question: 25-28

a_1	b_1	$a_1 + b_1 + c_1$	$=$	a_1	b_1	c_1
a_2	b_2	$a_2 + b_2 + c_2$		a_2	b_2	c_2
a_3	b_3	$a_3 + b_3 + c_3$		a_3	b_3	c_3

Taking RHS.

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow \begin{vmatrix} a_1 & b_1 & b_1+c_1 \\ a_2 & b_2 & b_2+c_2 \\ a_3 & b_3 & b_3+c_3 \end{vmatrix} \begin{vmatrix} c_3+c_2 \\ \\ \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & b_1 & a_1+b_1+c_1 \\ a_2 & b_2 & a_2+b_2+c_2 \\ a_3 & b_3 & a_3+b_3+c_3 \end{vmatrix} \begin{vmatrix} c_3+c_1 \\ \\ \end{vmatrix}$$

Hence LHS = RHS.

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$$\begin{vmatrix} a_1+t & a_2+t & a_3+t \\ a_1t+b_1 & a_2t+b_2 & a_3t+b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} - (1-t^2) \begin{vmatrix} a_1 & b_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Taking LHS.

$$= \begin{vmatrix} a_1(1-t^2) & a_2(1-t^2) & a_3(1-t^2) \\ a_1t+b_1 & a_2t+b_2 & a_3t+b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{vmatrix} R_1-tR_2 \\ R_1-tR_2 \\ R_1-tR_2 \end{vmatrix}$$

$$= (1-t^2) \begin{vmatrix} a_1 & a_2 & a_3 \\ a_1t+b_1 & a_2t+b_2 & a_3t+b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{vmatrix} a_1-a_1t^2 \\ a_1(1-t^2) \\ \end{vmatrix}$$

	a_1	a_2	a_3		$R_2 - tR_1$
$-(1-t^2)$	b_1	b_2	b_3	$R_2 - tR_1$	$a_1t + b_1 - a_1t$
	c_1	c_2	c_3		b_1

Hence LHS = RHS.

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$a_1 + b_1$	$a_1 - b_1$	c_1		a_1	b_1	c_1
$a_2 + b_2$	$a_2 - b_2$	c_2	$= -2$	a_2	b_2	c_2
$a_3 + b_3$	$a_3 - b_3$	c_3		a_3	b_3	c_3

Taking LHS					$c_1 + c_2$
	$2a_1$	$a_1 - b_1$	c_1	$c_1 + c_2$	$a_1 + b_1 + a_1 - b_1$
$=$	$2a_2$	$a_2 - b_2$	c_2		$2a_1$
	$2a_3$	$a_3 - b_3$	c_3		

-2	a_1	$a_1 - b_1$	c_1	
	a_2	$a_2 - b_2$	c_2	
	a_3	$a_3 - b_3$	c_3	

$= 2$	a_1	$-b_1$	c_1	$c_2 - c_1$
	a_2	$-b_2$	c_2	
	a_3	$-b_3$	c_3	

$= -2$	a_1	b_1	c_1	
	a_2	b_2	c_2	
	a_3	b_3	c_3	

Hence LHS = RHS.

28 -

$$\begin{array}{ccc|ccc}
 a_1 & b_1 + ta_1 & c_1 + \delta b_1 + sa_1 & & a_1 & a_2 & a_3 \\
 a_2 & b_2 + ta_2 & c_2 + \delta b_2 + sa_2 & \Rightarrow & b_1 & b_2 & b_3 \\
 a_3 & b_3 + ta_3 & c_3 + \delta b_3 + sa_3 & & c_1 & c_2 & c_3
 \end{array}$$

Taking LHS.

Transpose -

$$= \begin{array}{ccc}
 a_1 & a_2 & a_3 \\
 b_1 + ta_1 & b_2 + ta_2 & b_3 + ta_3 \\
 c_1 + \delta b_1 + sa_1 & c_2 + \delta b_2 + sa_2 & c_3 + \delta b_3 + sa_3
 \end{array}$$

$$= \begin{array}{ccc}
 a_1 & a_2 & a_3 \\
 b_1 & b_2 & b_3 \\
 c_1 + \delta b_1 + sa_1 & c_2 + \delta b_2 + sa_2 & c_3 + \delta b_3 + sa_3
 \end{array} \quad R_2 - tR_1$$

$$= \begin{array}{ccc}
 a_1 & a_2 & a_3 \\
 b_1 & b_2 & b_3 \\
 c_1 + sa_1 & c_2 + sa_2 & c_3 + sa_3
 \end{array} \quad R_3 - \delta R_2$$

$$= \begin{array}{ccc}
 a_1 & a_2 & a_3 \\
 b_1 & b_2 & b_3 \\
 c_1 & c_2 & c_3
 \end{array} \quad R_3 - sR_1$$

Hence LHS = RHS.

Question: 29-30

Show that $\det(A) = 0$ without directly evaluating.

29-

$$A = \begin{bmatrix} -2 & 8 & 1 & 4 \\ 3 & 2 & 5 & 1 \\ 4 & 10 & 6 & 5 \\ 4 & -6 & 4 & -3 \end{bmatrix}$$

As C_2 and C_4 are proportional so
 $\det(A) = 0$

Note: C_2 is obtained
 by xplying C_4
 with 2.

30

$$A = \begin{bmatrix} -4 & 1 & 1 & 1 & 1 \\ 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & -4 \end{bmatrix} \quad R_1 \rightarrow R_2 + R_3 + R_4 + R_5$$

Since one row become 0

$$\text{So } \det(A) = 0$$

Question: 31-32

It can be Proved that if a square matrix M is partitioned into block Triangular form as

$$M = \begin{bmatrix} A & 0 \\ C & B \end{bmatrix} \quad \text{or} \quad M = \begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$$

in which A & B are square, then
 $\det(M) = \det(A)\det(B)$. Use this to compute determinants of matrices in exercise 31 and 32

$$M = \begin{bmatrix} 1 & 2 & 0 & 8 & 6 & -9 \\ 2 & 5 & 0 & 4 & 7 & 5 \\ -1 & 3 & 2 & 6 & 9 & -2 \\ \hline 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & -3 & 8 & -4 \end{bmatrix}$$

Let

$$M = \begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$$

Compute A

$$\det(A) = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 5 & 0 \\ -1 & 3 & 2 \end{vmatrix}$$

expand C_3

$$\det(A) = 0 + 0 + (-1)^{3+3} (2) \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix}$$

$$\det(A) = -2(5-4)$$

$$\det(A) = 2$$

Compute B

$$\det(B) = \begin{vmatrix} 3 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 8 & -4 \end{vmatrix}$$

2nd Method

As B is Lower
Triangular
Matrix so

$$\det(B) = (3)(1)(-4)$$

$$\det(B) = (-1)^{3+1} (3) \begin{vmatrix} 1 & 0 \\ 8 & -4 \end{vmatrix} + 0 + 0 \quad \det(B) = -12$$

$$\det(B) = 3(-4-0)$$

$$\det(B) = -12$$

Now:

we don't need to compute determinant of C as its get multiplied by 0 and we are also give

$$\det(M) = \det(A)\det(B)$$

$$\det(M) = \begin{vmatrix} 2 & C \\ 0 & -12 \end{vmatrix}$$

$$\det(M) = -24 - 0$$

$$\det(M) = -24 \text{ Ans}$$

32-

$$M = \begin{vmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \\ 2 & 0 & 0 & 0 & 1 \end{vmatrix}$$

Let

$$M = \begin{bmatrix} A & 0 \\ C & B \end{bmatrix}$$

compute A

$$\det(A) = \begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix}$$

As A is upper triangular matrix:

$$\det(A) = (1)(1)(1)$$

$$\det(A) = 1$$

$$\det(M) = \begin{vmatrix} 1 & & \\ & 1 & \\ & & 1 \end{vmatrix}$$

Compute B

$$\det(B) = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix}$$

$$\det(B) = 1$$

$$\det(M) = \begin{vmatrix} 1 & 0 \\ c & 1 \end{vmatrix}$$

$$\det(M) = 1 \text{ Ans}$$

Question: 34

Find determinant of following matrix

$$\begin{vmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{vmatrix}$$

$$\det(A) = \begin{vmatrix} a+3b & a+3b & a+3b & a+3b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{vmatrix} \quad \begin{array}{l} R_1 \rightarrow R_1 + R_2 \\ R_3 + R_4 \end{array}$$

$$\det(A) = a+3b \begin{vmatrix} 1 & 1 & 1 & 1 \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{vmatrix}$$

$$\det(A) = (a+3b) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & a-b & 0 & 0 \\ 0 & 0 & a-b & 0 \\ 0 & 0 & 0 & a-b \end{vmatrix} \quad \begin{array}{l} R_2 - bR_1 \\ R_3 - bR_1 \\ R_4 - bR_1 \end{array}$$

Since it becomes upper triangular matrix so

$$\det(A) = (a+3b)(1)(a-b)(a-b)(a-b)$$

$$\det(A) = (a+3b)(a-b)^3$$

↳ Ans